
EXERCISE

Lecture 7: Auction Activity

AI508 — Algorithmic Game Theory

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1 Recall

Revenue maximization

$$\text{Rev}(r) = r(1 - F(r))$$

$$\frac{d}{dr} \text{Rev}(r) = 1 - rf(x) - F(x) \implies r^* = \frac{1 - f(r^*)}{F(r^*)}$$

We then derived ϕ s.t

$$\phi(v) = v - \frac{1 - F(v)}{f(v)} = \frac{\mathbb{P}(V \geq v)}{f(v)} \quad (\text{virtual welfare})$$

This means that

$$\phi(r^*) = 0, \text{ and } \phi(v) \leq v$$

2 cont. Revenue maximization

By the definition of ϕ :

$$\begin{aligned} \frac{d}{dr} \text{Rev}(r) &= -f(r)\phi(r) \\ \int_{r^*}^{\bar{v}} \frac{d}{dr} \text{Rev}(r) dr &= - \int_{r^*}^{\bar{v}} f(r)\phi(r) dr \\ \text{Rev}(\bar{v}) - \text{Rev}(r^*) &= - \int_{r^*}^{\bar{v}} f(r)\phi(r) dr \\ \text{Rev}(r^*) &= \int_{r^*}^{\bar{v}} f(r)\phi(r) dr \\ \int_0^{\bar{v}} \phi(r) \mathbb{1}\{v \leq r^*\} f(r) dr &= \mathbb{E}[\phi(r) \mathbb{1}\{v \leq r^*\}] \quad (\text{Zeroes the integral below } r^*) \end{aligned}$$

Given the allocation rule

$$X_{r^*}(v) = \begin{cases} 1 & \text{if } v \geq r^* \\ 0 & \text{otherwise} \end{cases}$$

Then

$$\text{Rev}(r^*) = \mathbb{E}_{v \sim F}[\phi(v)X_{r^*}(v)]$$

Take (x, p) from Myerson's Lemma

$$\mathbb{E}_{v_i \sim F_i}[p_i(\vec{v})] = \mathbb{E}_{v_i \sim F_i}[\phi_i(v_i)X_i(\vec{v})] \quad \forall i \quad \forall v_{-i}$$

Then,

$$\mathbb{E}_{\vec{v} \sim F} \left[\sum_{i=1}^n p_i(\vec{v}) \right] = \mathbb{E}_{\vec{v} \sim F} \left[\sum_{i=1}^n \phi_i(v_i) X_i(\vec{v}) \right]$$

We now have that

$$\text{Revenue Maximization} = \arg \max_x \mathbb{E}_{v \sim F} \left[\sum \phi_i(v_i) x_i(\vec{v}) \right]$$

This only work when $\phi(v) \geq 0$ and when ϕ is monotone. Add the following constraint.

Given that $v_1 < v_2$

$$\phi(v_1) \leq \phi(v_2)$$

$$r^* = \phi^{-1}(0)$$

3 Exercises

1 Consider a single-item auction with at least three bidders. The highest bidder receives the item but pays the *third*-highest bid. Show that truthful bidding, is not a dominant strategy.

2 A seller has k identical items and $n > k$ bidders. Each bidder wants at most one item. Design a generalization of the second-price auction for this setting. Specify its allocation and payment rules, and prove that truthful bidding, is a dominant strategy

For allocation

$$X_i = \begin{cases} 1 & \text{if } b_i \in \text{topk}(B) \\ 0 & \text{otherwise} \end{cases}$$

then for $b_1 \geq \dots b_k \geq b_{k+1}$, all k players pay b_{k+1} .

For allocation

$$X_i = \begin{cases} 1 & \text{if } b_i \leq b_{-i} \\ 0 & \text{otherwise} \end{cases}$$

then for $b_1 \leq \dots b_k$, client pay b_2 .